

Bank

Customer Attrition Analysis



Problem Statement

What factors are associated with higher customer attrition?

Identify key differences between customers who churned and those who stayed. Help the bank understand what types of customers are more likely to churn so they can take preventive actions.

Data Set Overview

<https://www.kaggle.com/datasets/teralasowmya/bankchurner>

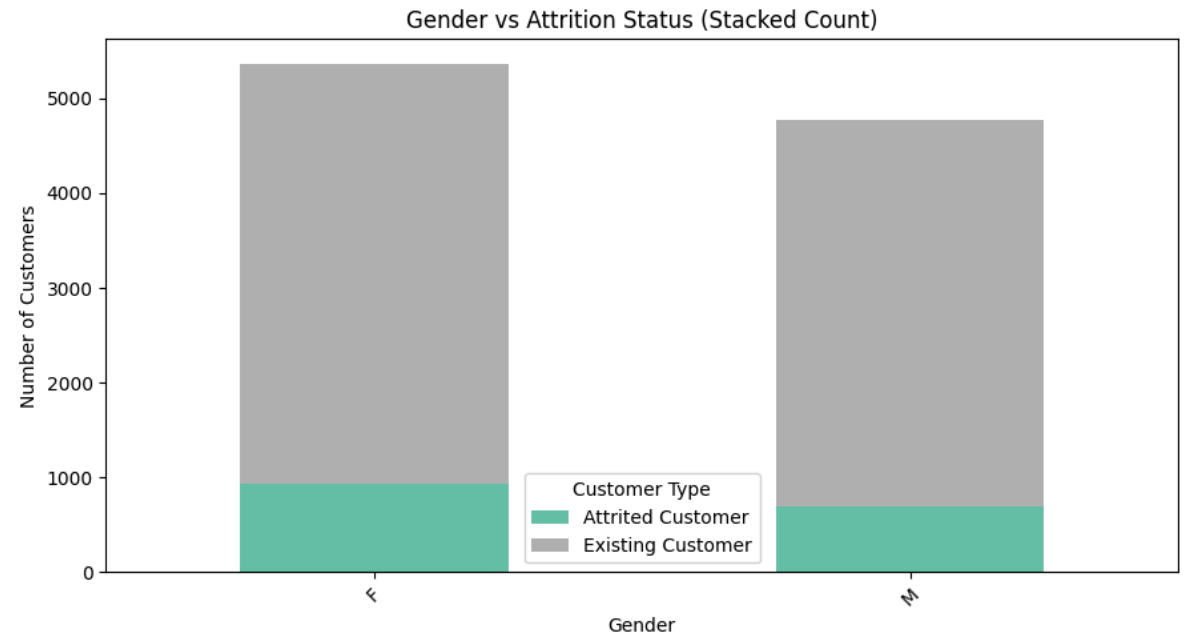
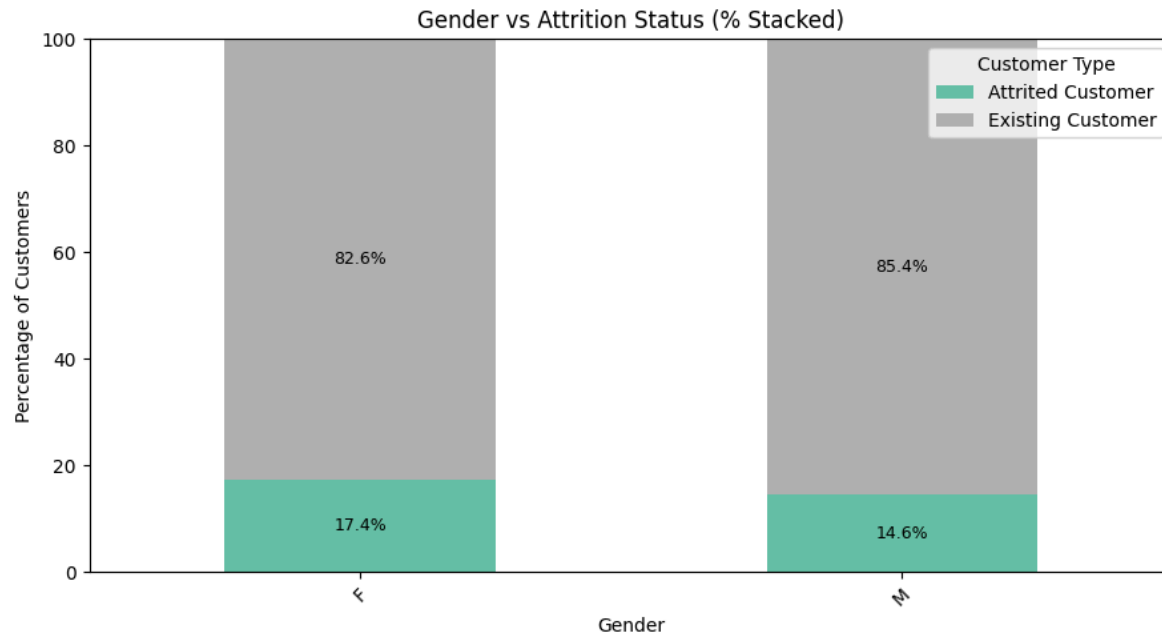
Features	Data Types	Description
CLIENTNUM	int64	Client ID
Attrition_Flag	object	If account is closed "Attrited Customer" else "Existing Customer"
Customer_Age	int64	Age of customer in years
Gender	object	Gender of customer
Dependent_count	int64	No. of dependents of customer : 0,1,2,13,4,5
Education_Level	object	High School, Graduate, Uneducated, Unknown, College, Post-Graduate, Doctorate
Marital_Status	object	Married, Single, Unknown , Divorced
Income_Category	object	Income Categories between \$40k to over \$120k
Card_Category	object	Blue, Silver, Gold, Platinum
Months_on_book	int64	No of months has been a customer
Total_Relationship_Count	int64	Total no. of products held by the customer
Months_Inactive_12_mon	int64	No. of months inactive in the last 12 months
Contacts_Count_12_mon	int64	No. of Contacts between the customer and bank in the last 12 months
Credit_Limit	float64	Credit limit on credit card
Total_Revolving_Bal	int64	The balance that carries over from one month to the next is the revolving balance
Avg_Open_To_Buy	float64	Open to Buy refers to the amount left on the credit card to use(Average of last 12 months)
Total_Amt_Chng_Q4_Q1	float64	Ratio of the total transaction amount in 4th quarter and the total transaction amount in 1st quarter
Total_Trans_Amt	int64	Total Transaction Amount
Total_Trans_Ct	int64	Total Transaction Count
Total_Ct_Chng_Q4_Q1	float64	Ratio of the total transaction count in 4th quarter and the total transaction count in 1st quarter
Avg_Utilization_Ratio	float64	Represents how much of the available credit the customer spent



Exploratory Data Analysis

Distribution by Gender

Attrition rate high among female customers.



17.4%

ATTRITION RATE

FEMALE

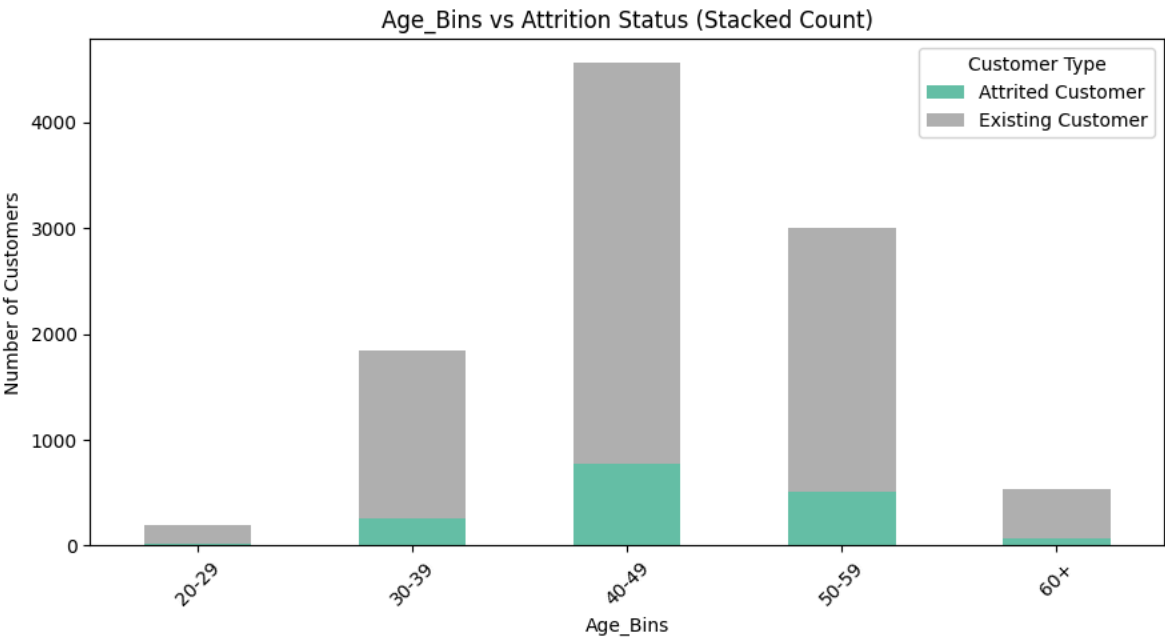
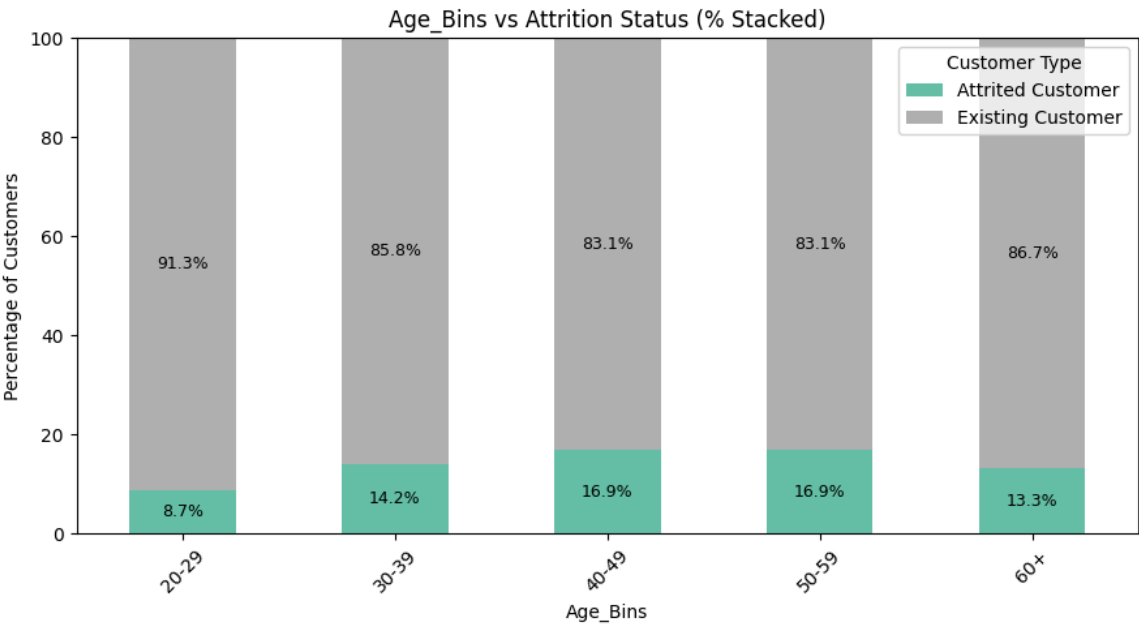
14.6%

ATTRITION RATE

MALE

Distribution by Age

Attrition high amongst customers aged group 50-59 yrs and 40-49 yrs.



16.9%

ATTRITION RATE

50-59 YRS

16.9%

ATTRITION RATE

40-49 YRS

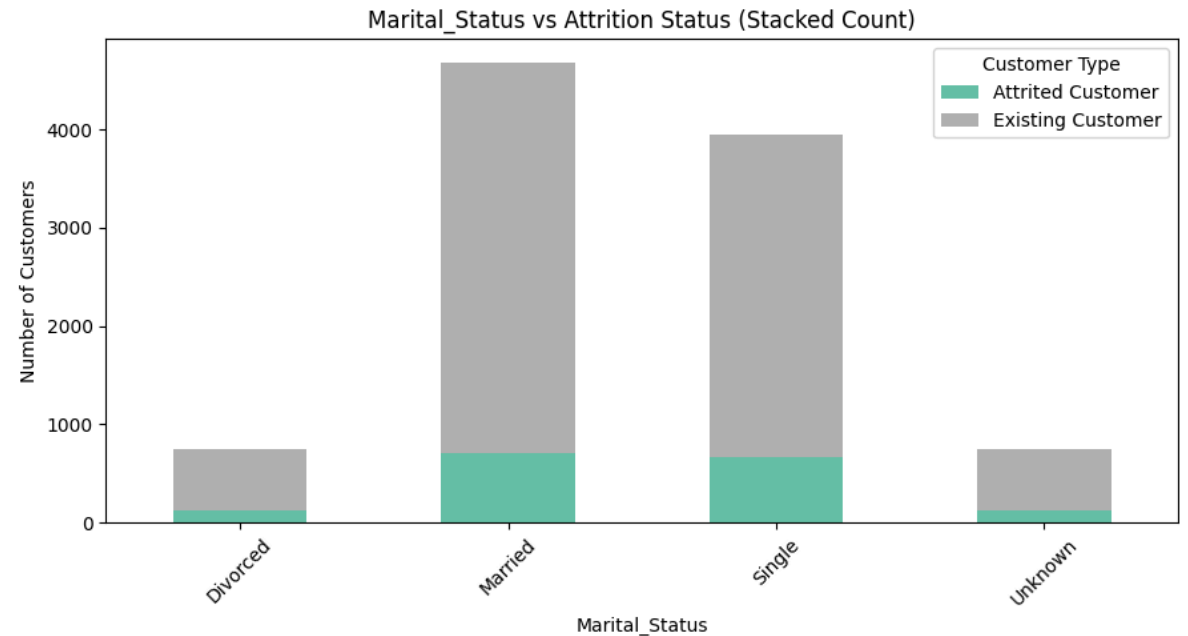
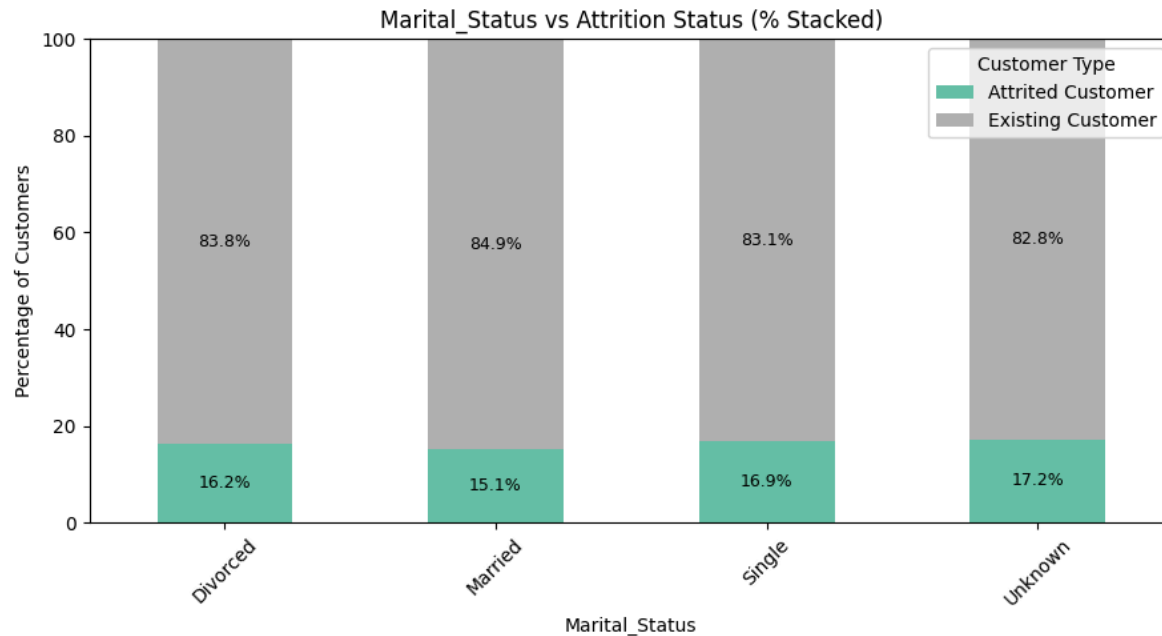
14.2%

ATTRITION RATE

30-39 YRS

Distribution by Marital Status

Attrition rate high amongst single and divorced customers.



16.9%

ATTRITION RATE

SINGLE

16.2%

ATTRITION RATE

DIVORCED

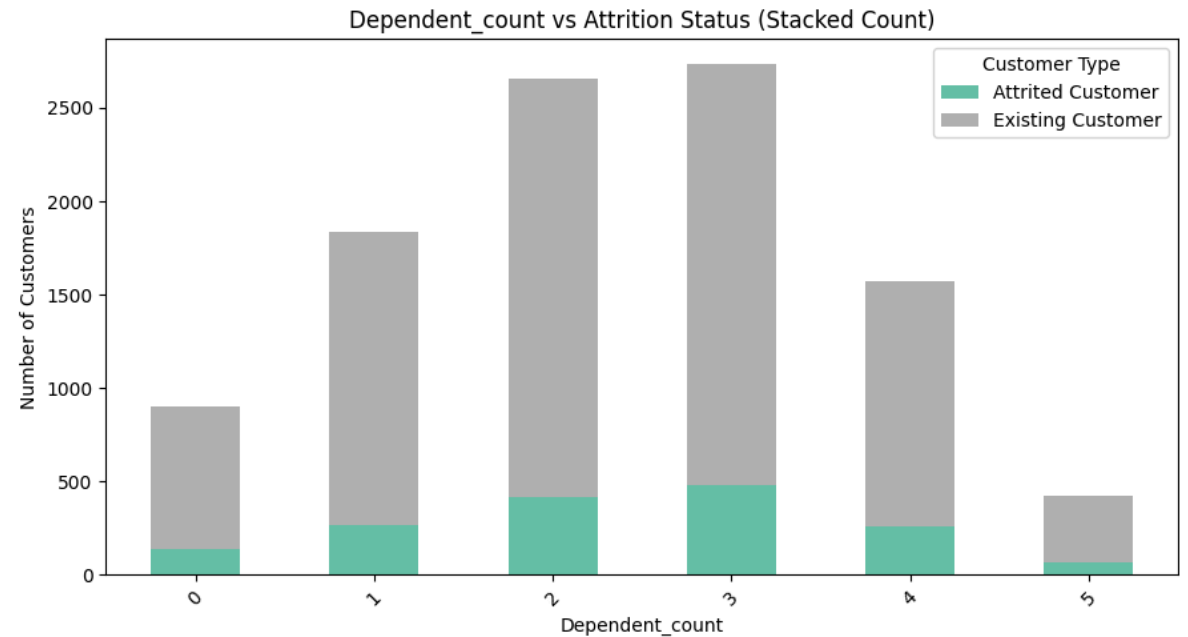
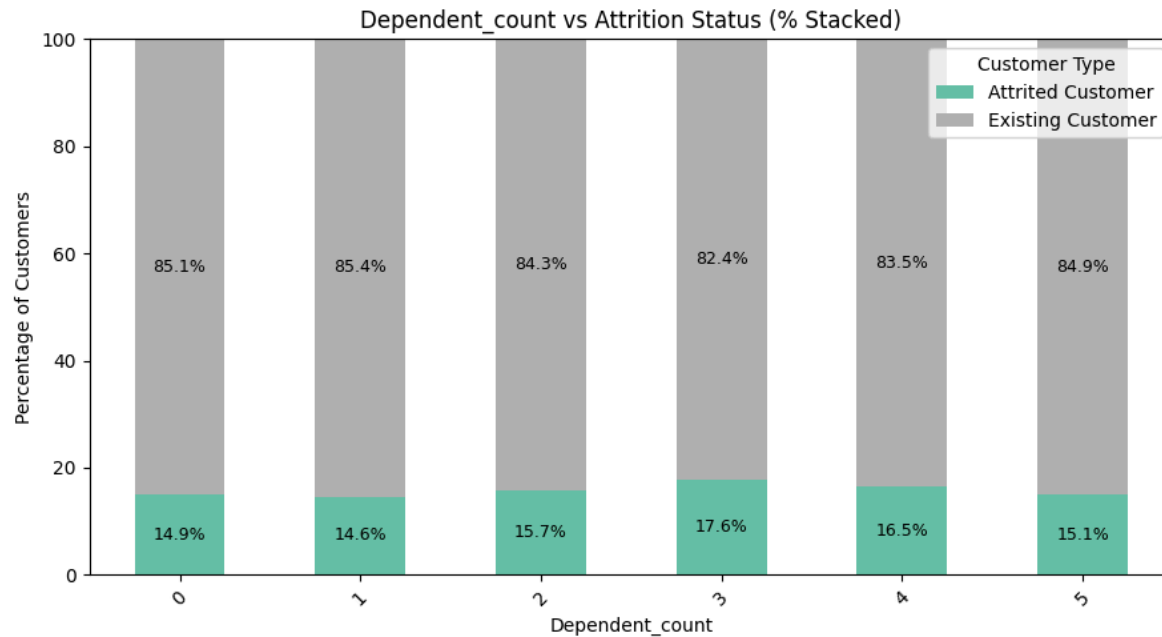
15.1%

ATTRITION RATE

MARRIED

Distribution by Dependent Count

Attrition high amongst customers with dependent count of three, four and two.



17.6%

ATTRITION RATE

3 DEPENDENT

16.5%

ATTRITION RATE

4 DEPENDENT

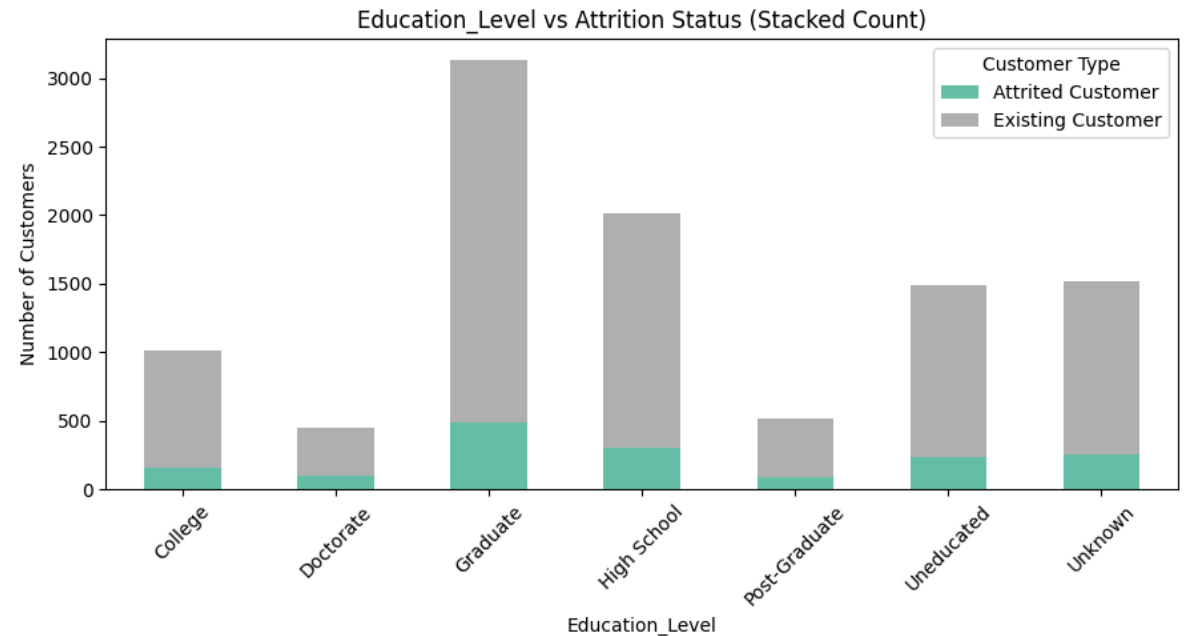
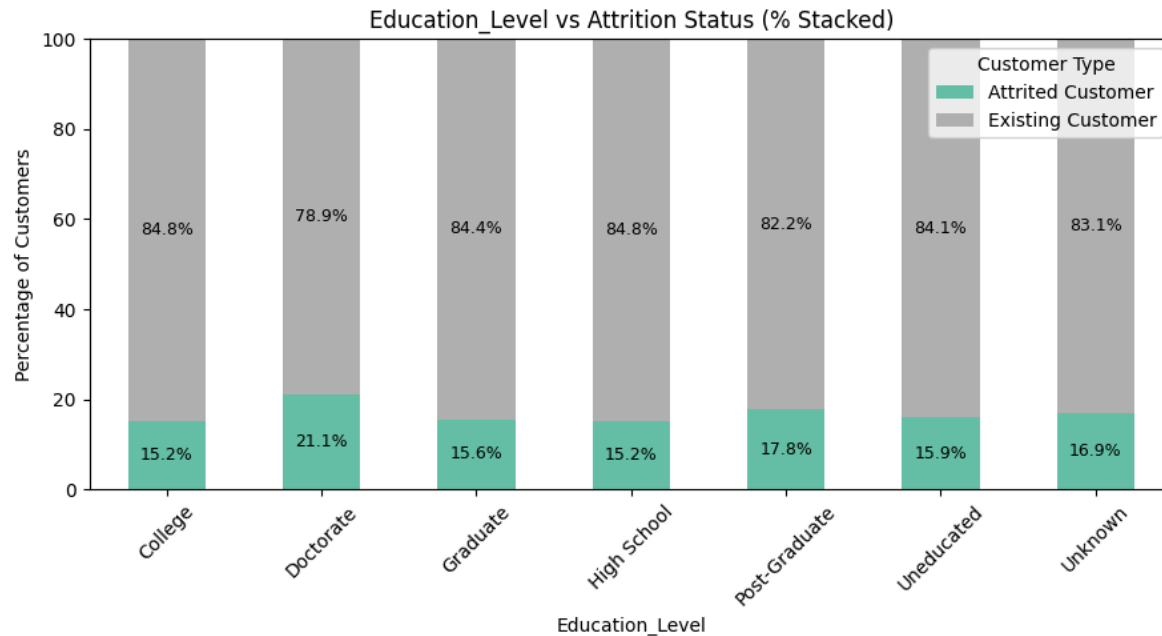
15.7%

ATTRITION RATE

2 DEPENDENT

Distribution by Education Level

Attrition rate highest amongst those categories with a high count - Uneducated, Graduate and High School.



15.9%

ATTRITION RATE

UNEDUCATED

15.6%

ATTRITION RATE

GRADUATE

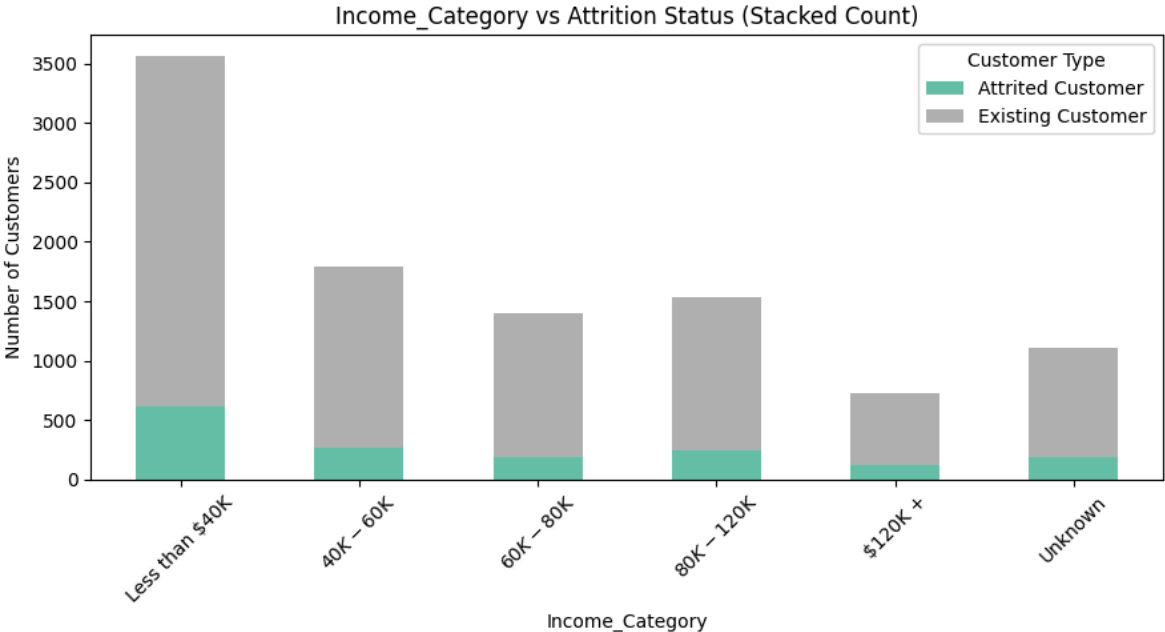
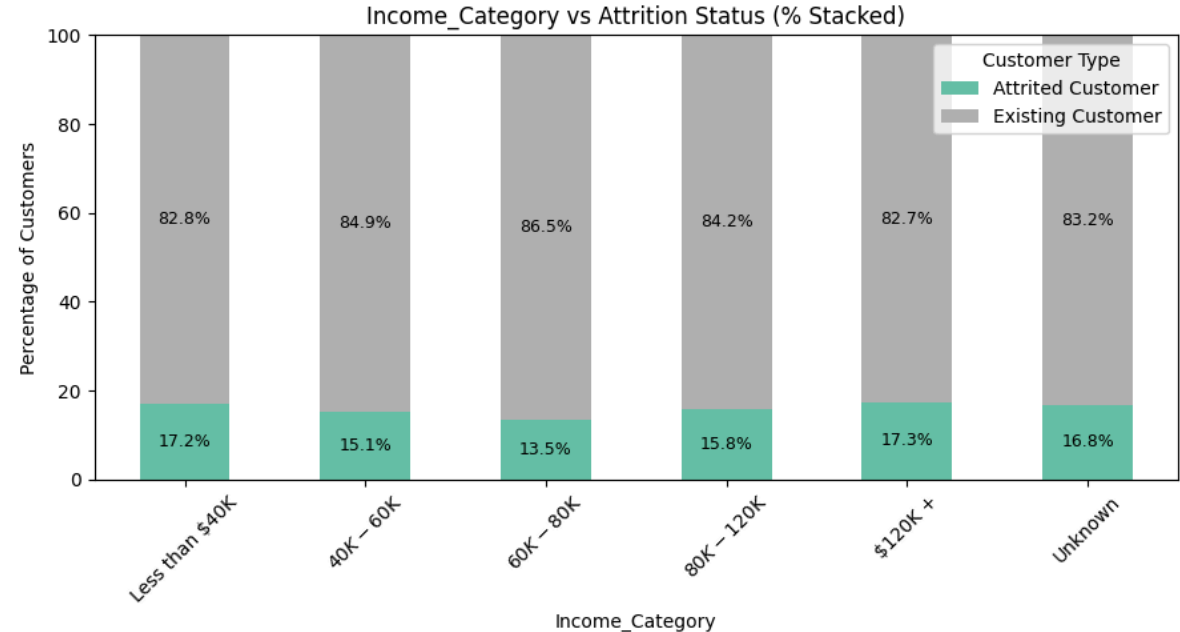
15.2%

ATTRITION RATE

HIGH SCHOOL

Distribution by Income Category

Attrition rate highest amongst those categories with a high count - Less than \$40k, \$40-\$60k and \$80-\$120k.



17.2%

ATTRITION RATE

< \$40K

15.1%

ATTRITION RATE

\$40K - \$60K

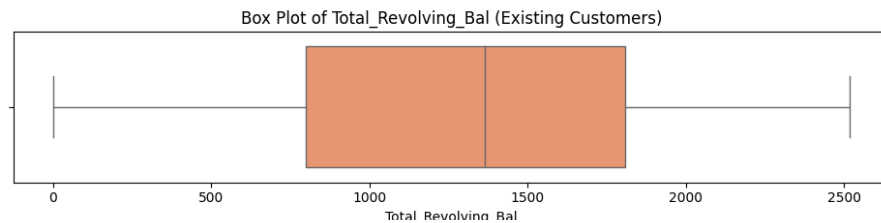
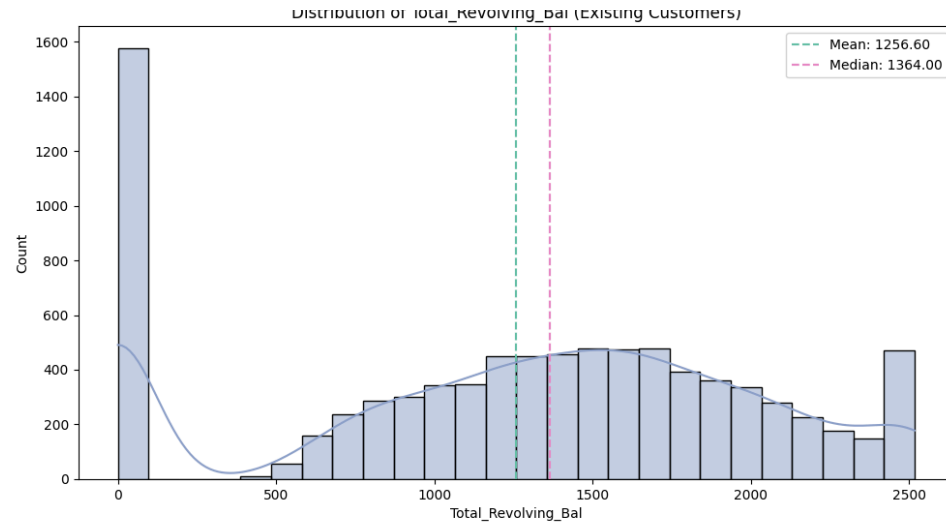
15.8%

ATTRITION RATE

\$80K - \$120K

Distribution of Total Revolving Balance

The mean & median for attrited customers are low compared to existing customers.



\$1257

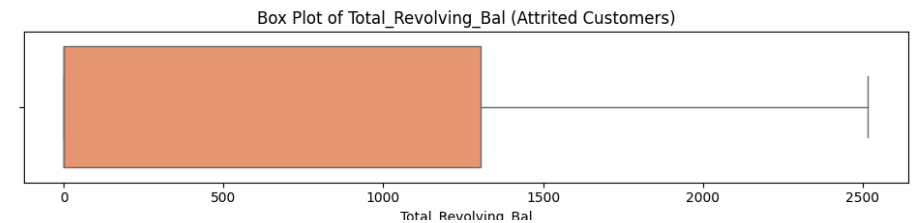
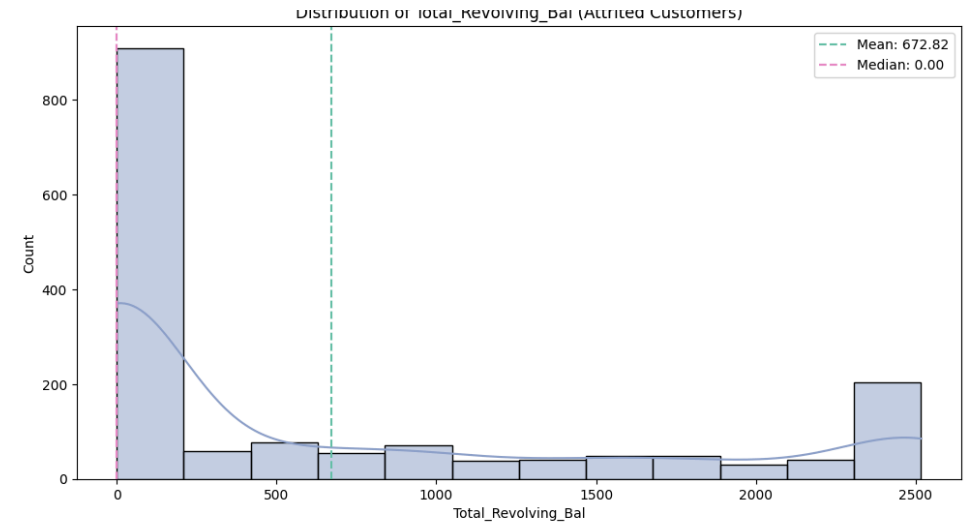
MEAN

EXISTING CUSTOMER

\$1364

MEDIAN

EXISTING CUSTOMER



\$673

MEAN

ATTRITED CUSTOMER

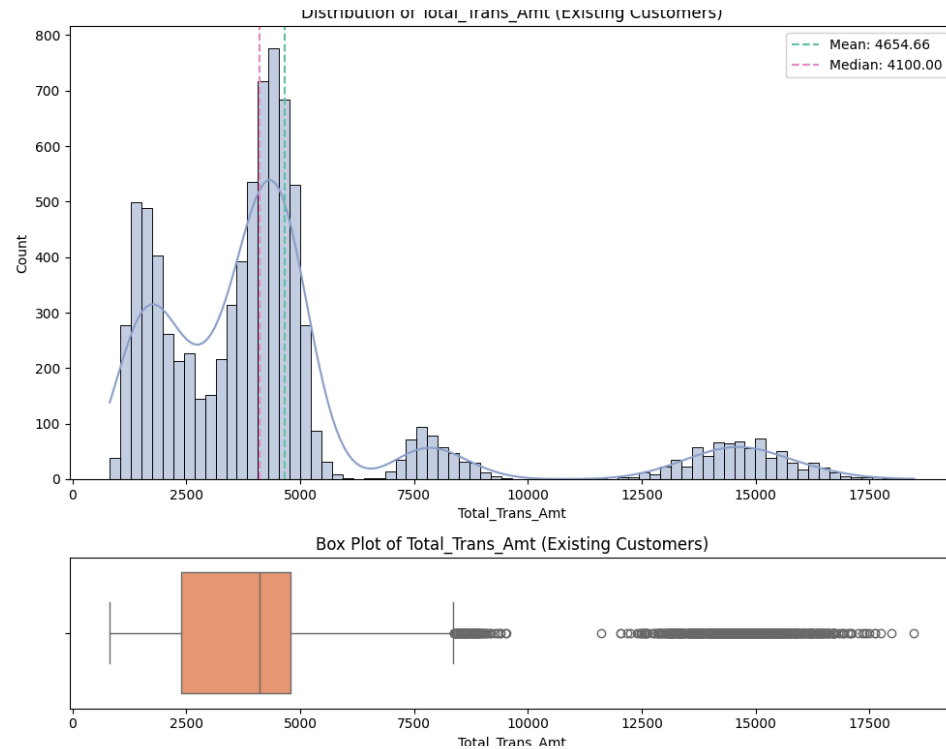
\$0

MEDIAN

ATTRITED CUSTOMER

Distribution of Total Transaction Amount

The mean & median for attrited customers are low compared to existing customers.



\$4655

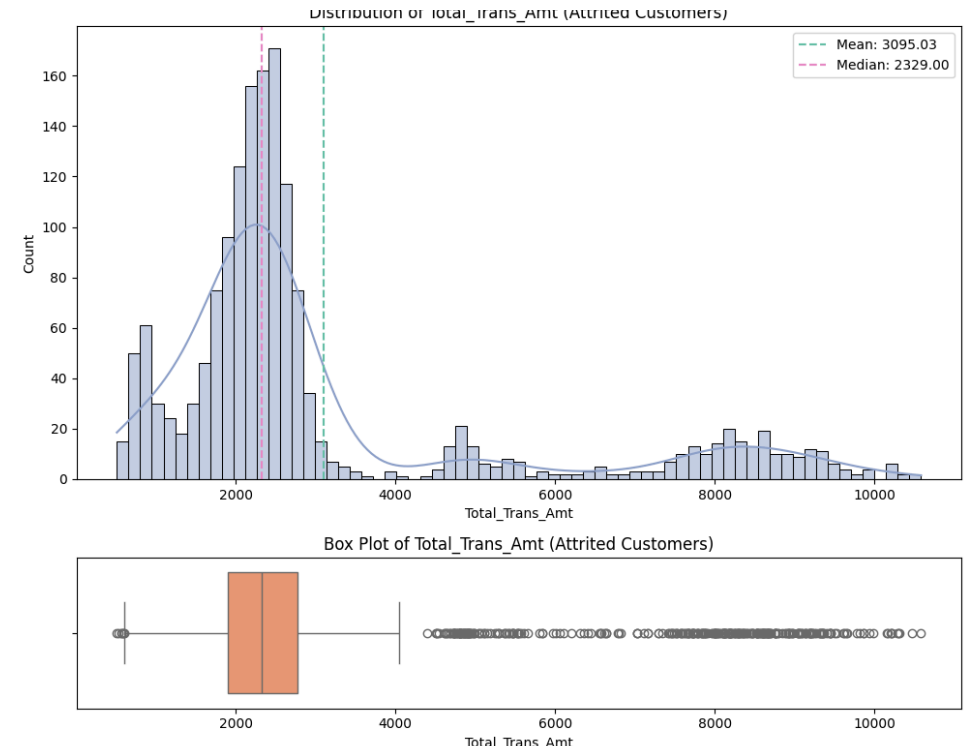
MEAN

EXISTING CUSTOMER

\$4100

MEDIAN

EXISTING CUSTOMER



\$3095

MEAN

ATTRITED CUSTOMER

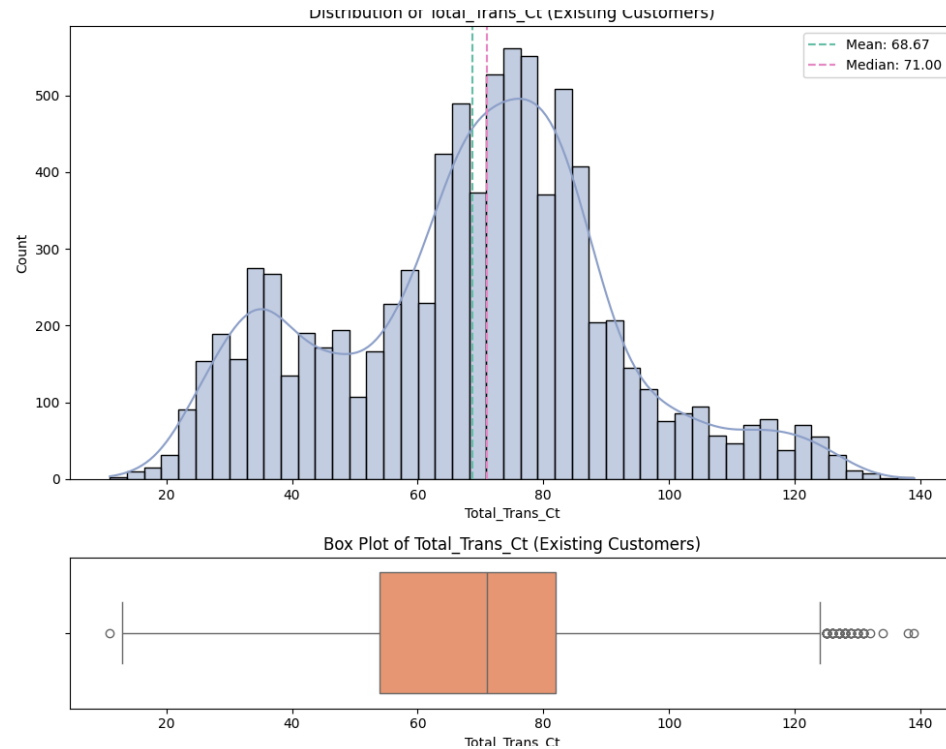
\$2329

MEDIAN

ATTRITED CUSTOMER

Distribution of Total Transaction Count

The mean & median for attrited customers are low compared to existing customers.



\$69

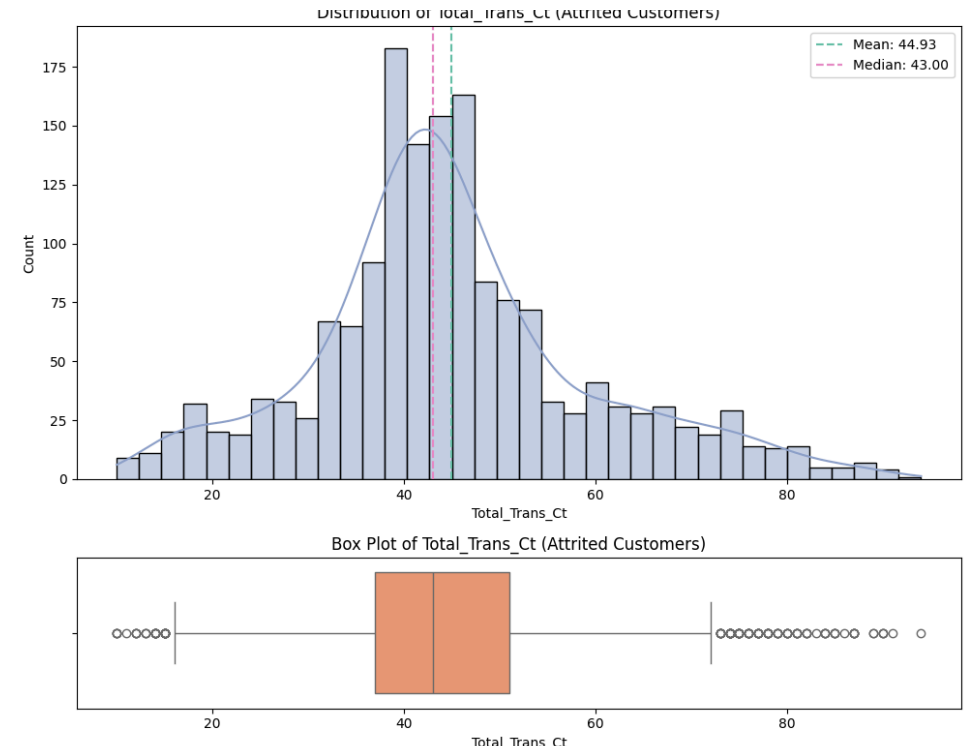
MEAN

EXISTING CUSTOMER

\$71

MEDIAN

EXISTING CUSTOMER



\$45

MEAN

ATTRITED CUSTOMER

\$43

MEDIAN

ATTRITED CUSTOMER



Hypothesis Testing

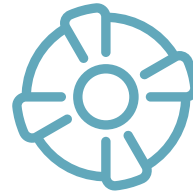
Numerical Columns

One tailed Two Sample independent T test



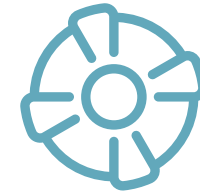
Total Revolving Balance

Null hypothesis rejected: The mean total revolving balance is less for attrited customers than for existing customers.



Total Transaction Amount

Null hypothesis rejected: The mean total transaction amount is less for attrited customers than for existing customers.



Total Transaction Count

Null hypothesis rejected: The mean total transaction count is less for attrited customers than for existing customers.

Categorical Columns

Chi Squared Test



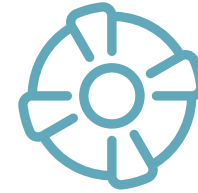
Gender

Null hypothesis rejected: Significant association with Attrition



Age Category

Null hypothesis rejected: Significant association with Attrition



Marital Status

Null hypothesis accepted: No significant association



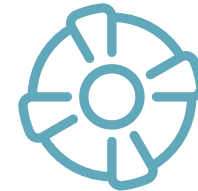
Dependent Count

Null hypothesis accepted: No significant association



Education Level

Null hypothesis accepted: No significant association



Income Category

Null hypothesis rejected: Significant association with Attrition



Business Insights and Recommendations

Insights and Recommendations

- **Distribution by Gender** - Attrition rate high among female customers.
Recommendation - *Create women centric focus groups to understand their pain points and provide financial planning tools.*
- **Distribution by Age** - Attrition high amongst customers aged group 50-59 yrs and 40-49 yrs.
Recommendation - *Conduct surveys or focus groups to understand their pain points. Provide mid-life financial planning tools, loyalty programs, or exclusive offers.*
- **Distribution by Income Category** - Attrition rate highest amongst those categories with a high count - Less than \$40k, \$40-\$60k and \$80-\$120k.
Recommendation - *For customers under the \$40K category, design tailored financial products, rewards, or engagement programs for this segment. Offer personalized outreach, financial literacy resources, or incentives to increase perceived value and customer loyalty.*
- **Total Revolving Balance** - The mean & median for attrited customers are low compared to existing customers.
- **Total Transaction Amount** - The mean & median for attrited customers are low compared to existing customers.
- **Total Transaction Count** - The mean & median for attrited customers are low compared to existing customers.
Recommendation - *Build churn prediction models using these indicators to identify at-risk customers. Trigger timely retention actions such as personalized communication, rewards, or relationship manager calls.*